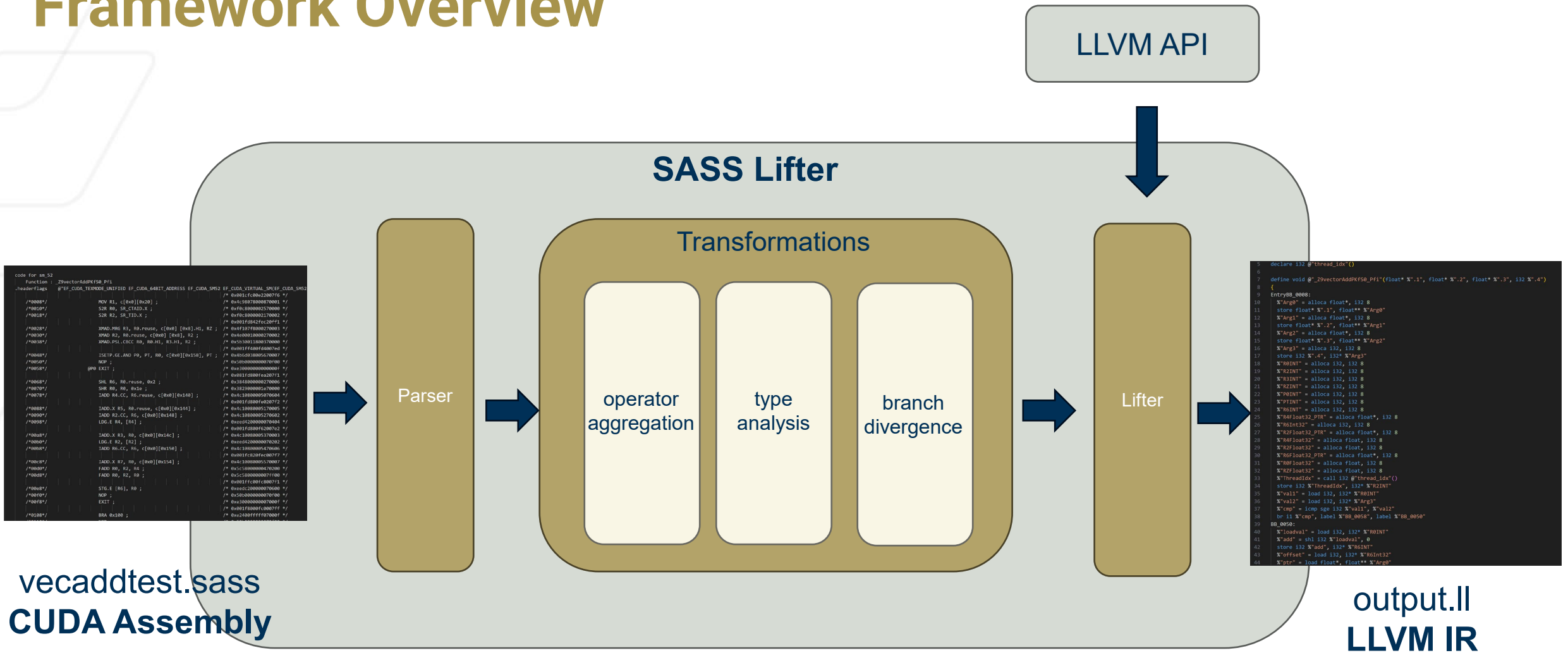
Project 5 Q&A

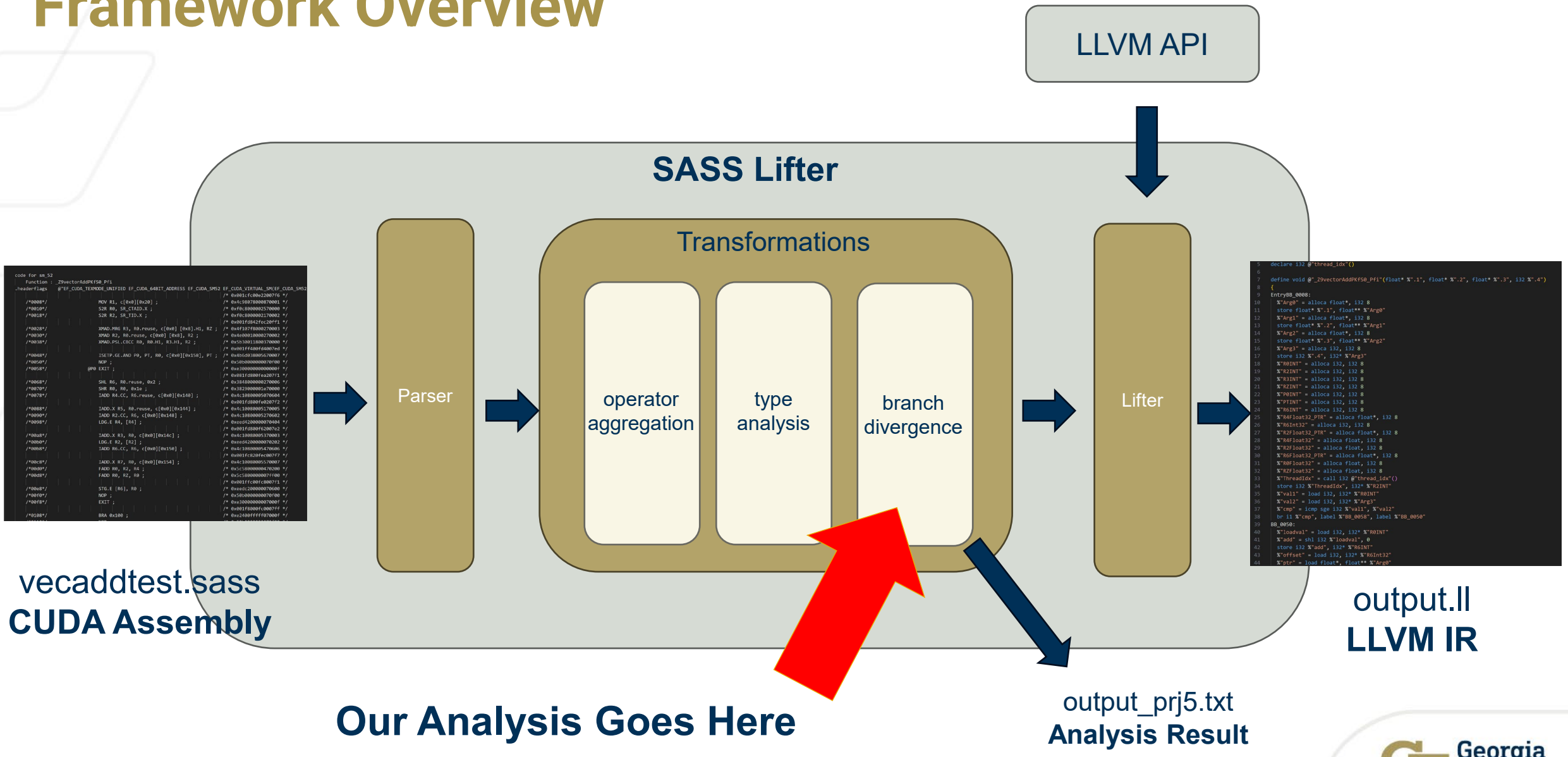
Warp Branch Divergence Detection

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Framework Overview



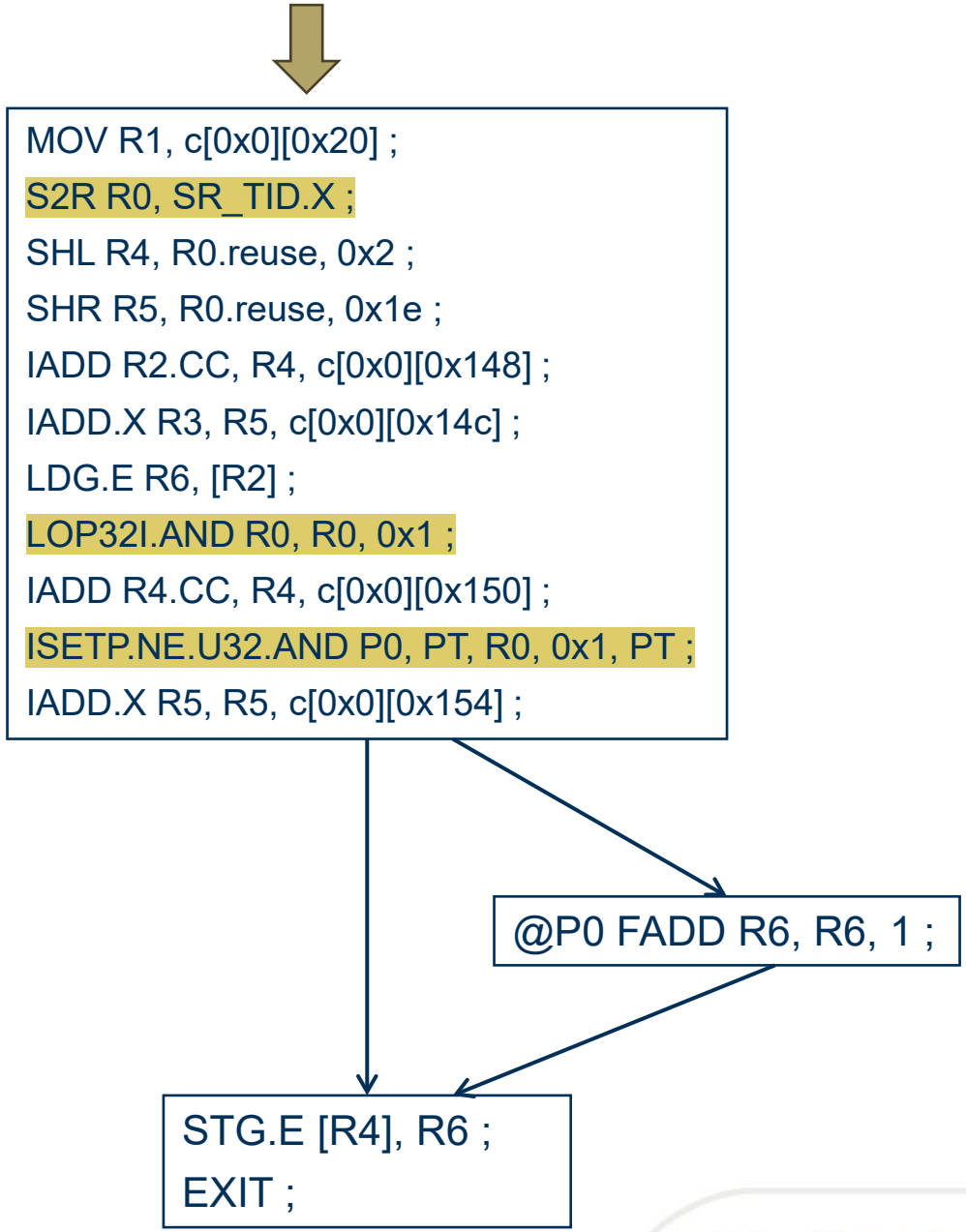
Framework Overview



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Example

- *S2R* accesses threadIdx.x and stores at R0
- *LOP32I.AND* redefines R0
- *ISETP.NE.U32.AND* uses R0
  - This is the only branch divergent instruction

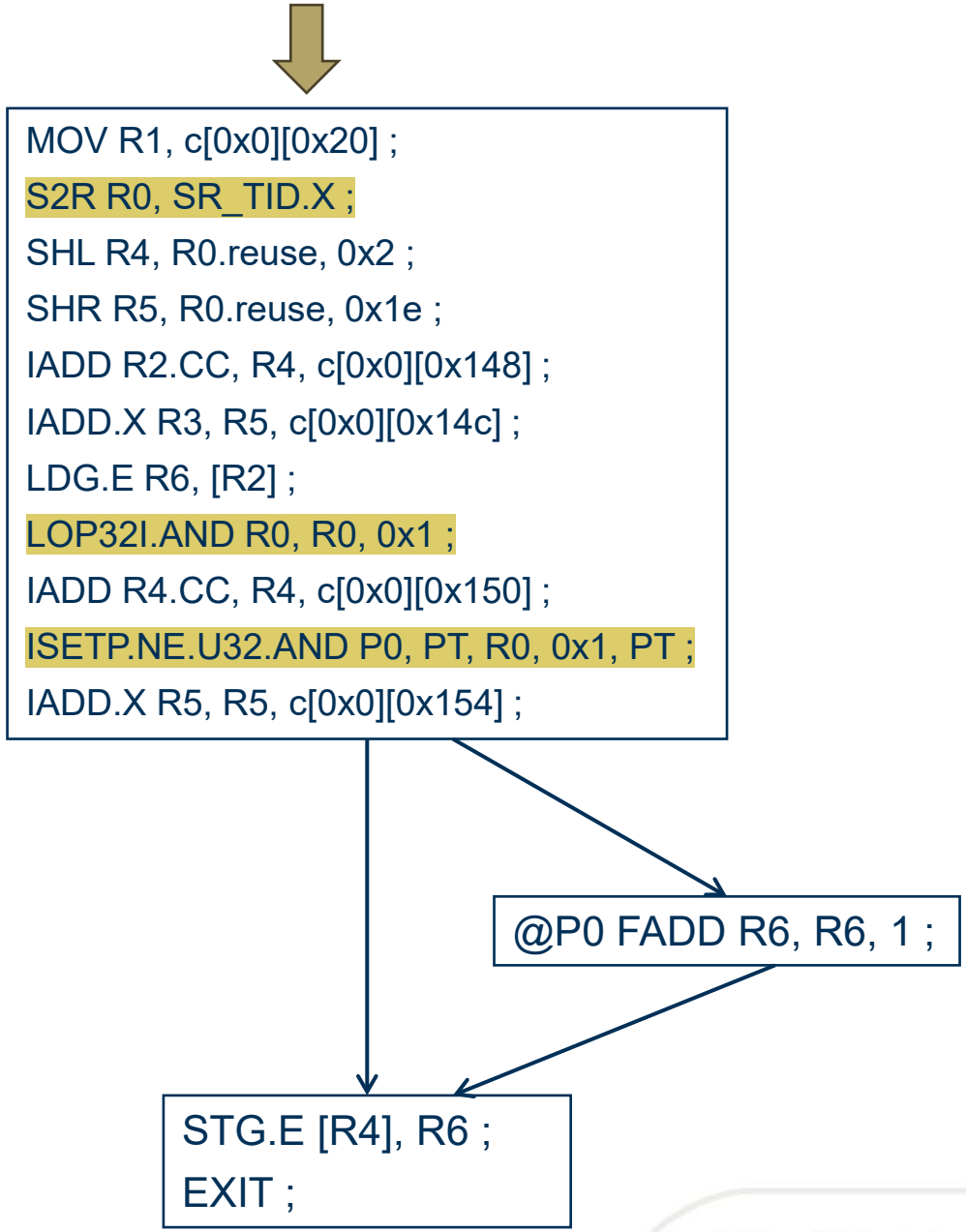


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Example

Analysis steps(for reference)

1. Find out reg-to-def relationships
2. Find out def-to-use relationships
3. Find out threadIdx instructions
4. Taint other instructions across def-use chains
5. Filter out branch instructions
6. Done!

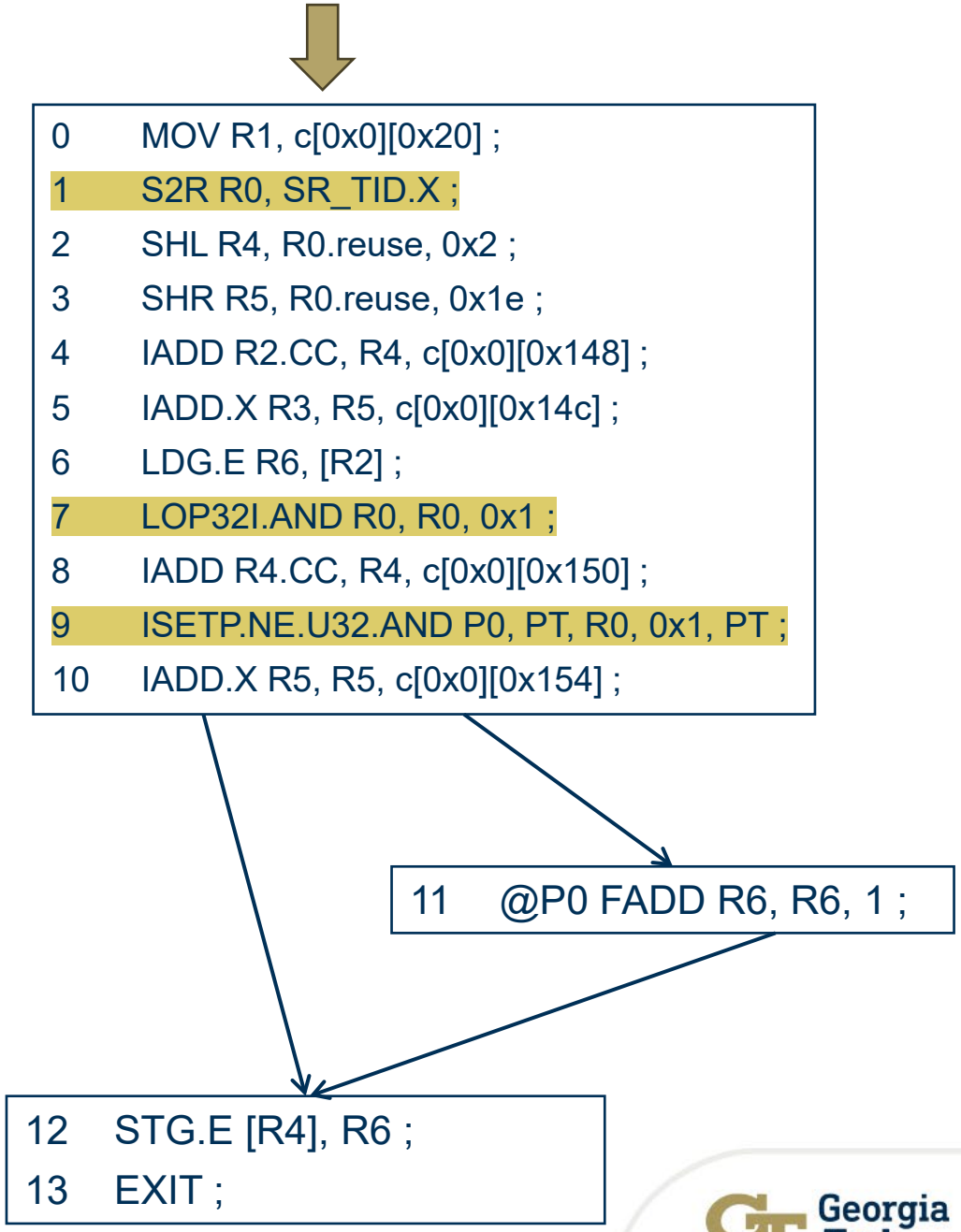


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Example

Constructing Reg to Def table

- At each instruction, what definition(s) are register(s) currently storing?
  - At #6, R0 is defined by #1
  - At #7, R0 is defined by #7
  - At #12, R0 is still defined by #7
    - Propagation across basic blocks needed
- A register can have multiple potential definitions.
  - At #12, R6 is defined by {#6, #11}
- If the cfg contains a loop, one propagation will not be enough
  - Keep updating until nothing changes

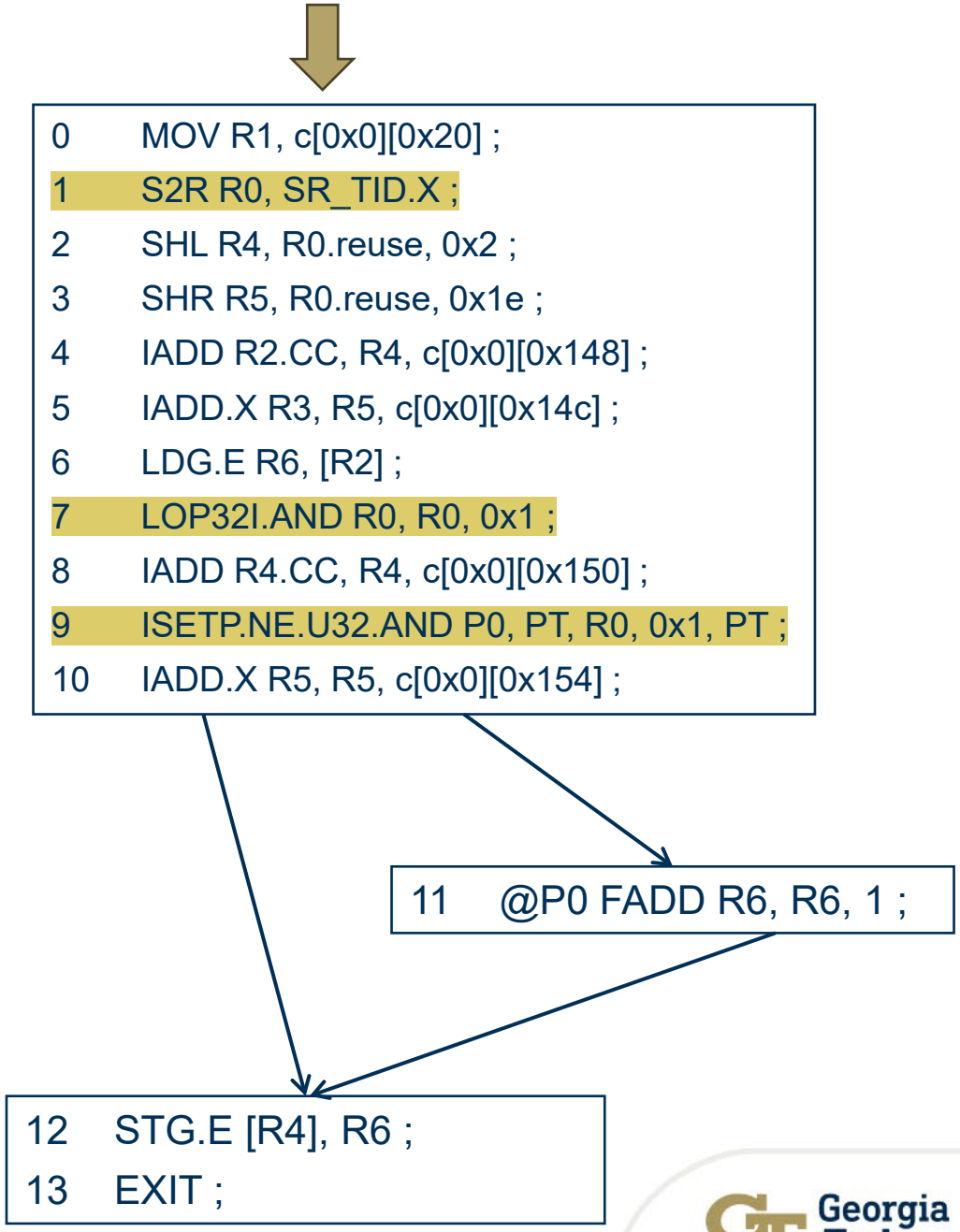


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Example

Constructing Def to Use table

- Similar strategy
- Propagate def-use info through instructions and basic blocks
- A definition can have multiple uses
- Rely on definition, not register names!
  - E.g. #12 uses #8, not #4
- Keep iterating until no change



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Tips

- CUDA assembly(SASS) has no official documentation, but many unofficial documentations are available online
- Loop-related test cases have source code provided, they can help you understand the assembly if you get stuck

Tips

- Be careful with designing data structures for reg-to-def and def-to-use tables.
  - Use Dict, Set, etc from Python
- Instructions like “ISETP.GE.AND” are set predicate instructions
  - The postfix allows chaining of boolean operations
  - ISETP.GE.AND P2, PT, R8, 0x100, PT is equivalent to P2 = (R8>0x100 & True);
  - **These are essentially branch instructions** because predicate is used for conditional execution